Clean Accuracy Explains Most of the Robustness Gap: A Controlled Comparison of ResNet-18 and VGG-16 under Adversarial Training on CIFAR-100

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***Abstract*—Residual networks are often reported to be more robust than plain sequential networks after adversarial training, but such comparisons rarely separate robustness from clean accuracy. We train ResNet-18 and VGG-16 on CIFAR-100 under an identical recipe, with and without a mixed clean/adversarial training objective, and evaluate them with FGSM, PGD-7, PGD-20 and AutoAttack at ℓ_∞ budgets up to $\epsilon = 8/255$, averaging over three random seeds. Adversarial training at $\epsilon = 4/255$ raises PGD-20 accuracy at that budget to 28.17% (ResNet-18) and 21.84% (VGG-16) (AutoAttack: 27.37% and 20.93%), at a clean-accuracy cost of 5.15 and 3.94 points, but transfers poorly to $\epsilon = 8/255$ (AutoAttack: 5.77% and 4.13%). ResNet-18 is ahead in absolute robust accuracy at every budget. However, when we decompose robust accuracy into clean accuracy times the fraction of correctly classified inputs that survive the attack, about 70% of the gap at $\epsilon \leq 4/255$ and all of it at $\epsilon = 8/255$ is explained by ResNet-18's higher clean accuracy; with a better-trained VGG-16 (single linear head) this share is still 45%. We also find that checkpoint selection on a small robust validation set adds substantial variance to the reported trade-off, and that the "failure cases are better learned" phenomenon recently reported for PGD-based adversarial training does not appear under our weaker objective, although adversarial training moves both networks towards it.**

***Index Terms*—Adversarial machine learning, adversarial training, robustness, accuracy-robustness trade-off, ResNet-18, VGG-16, CIFAR-100.**

I. Introduction

Convolutional Neural Networks (CNNs) reach high accuracy on image classification, yet their predictions can be changed by small, carefully crafted perturbations of the input [1], [2]. Such attacks also work in the physical world, e.g., stickers on a stop sign mislead a traffic-sign classifier [3], and concern any deployed model whose input an attacker can influence [4].

Many early defences only obfuscated gradients and were broken by adaptive attacks [4], [5]; adversarial training (AT) remains the most reliable empirical defence [6], [7]. AT has two well-known costs. First, it lowers accuracy on clean data, the accuracy-robustness trade-off [8]–[12]. Second, its outcome depends on the network architecture [13], [14].

These two effects are entangled whenever architectures are compared. A prediction can only remain correct under attack if it was correct on the clean input, so a model that classifies more clean inputs correctly also has more inputs that can survive an attack. Benchmarks usually rank models by absolute robust accuracy [13], [15], and the most robust network is typically also the most accurate one. It is then unclear whether the architecture makes the model harder to attack, or simply classifies more inputs correctly to begin with. The first would justify choosing an architecture for robustness; the second only means that a stronger classifier stays stronger.

We study this question in a controlled setting. We choose ResNet-18 [16] and VGG-16 [17] as two classical representatives of residual and plain sequential CNNs, and CIFAR-100 as a dataset on which clean accuracy is far from saturated, so that the two networks differ clearly in clean accuracy. Both networks are trained with exactly the same optimiser, schedule, data augmentation and mixed clean/adversarial AT objective, over three random seeds, and are evaluated with single-step, multi-step and ensemble attacks at three perturbation budgets. Fig. 1 gives an overview of the pipeline. Our contributions are:

- A multi-seed benchmark of a residual and a sequential CNN on CIFAR-100, with and without mixed-loss AT, under FGSM, PGD-7, PGD-20 and AutoAttack [18], with sanity checks against gradient masking.
- A simple decomposition of robust accuracy into clean accuracy and conditional robust accuracy, showing that a large part of ResNet-18's robustness advantage is inherited from its higher clean accuracy, also against a better-trained VGG-16 control.
- Two methodological observations: best-checkpoint selection on a small robust validation set makes the trade-off noisy, and the perception-consistency pattern of [12] does not appear under a weaker AT objective.

II. Related Work

A. Adversarial Attacks and Robustness Evaluation

Goodfellow et al. attributed adversarial vulnerability [1] to the locally linear behaviour of DNNs and proposed the single-step FGSM attack [2]; Madry et al. generalised it into the

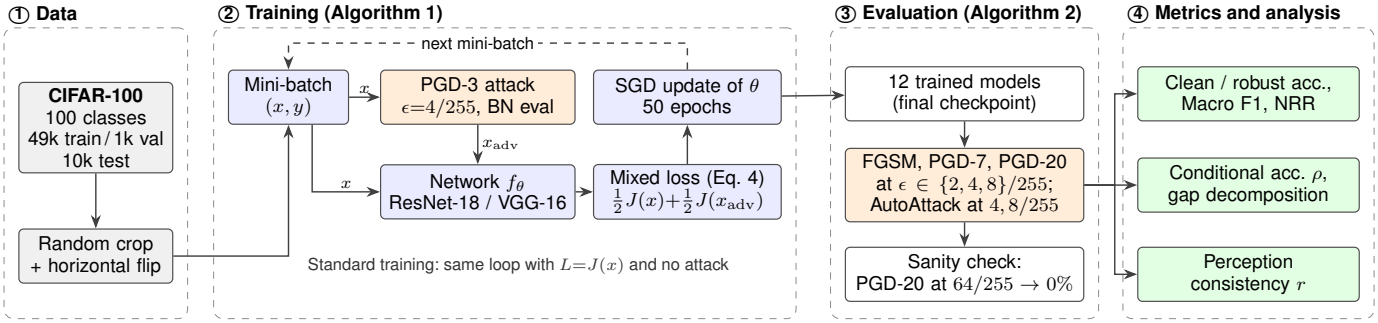


Fig. 1. Overall architecture of the study. ResNet-18 and VGG-16 are trained on CIFAR-100 with standard cross-entropy or with mixed-loss adversarial training on PGD-3 examples generated on the fly (Algorithm 1). The 12 trained models are then attacked and analysed (Algorithm 2).

multi-step PGD attack, which approximates the worst case inside an ℓ_p ball [6]. Because many early defences resisted only weak attacks [4], [5], evaluation has moved to stronger, standardised protocols: AutoAttack [18] combines several parameter-free attacks to avoid overestimating robustness, and RobustBench [15] tracks models evaluated in this way. Recent AT papers typically report clean accuracy, PGD-20 accuracy and AutoAttack accuracy averaged over several runs [12]; we follow the same protocol.

B. Adversarial Training and Its Variants

AT is formulated as a min-max problem in which the inner maximisation crafts adversarial examples and the outer minimisation fits the model to them [6]; Bai et al. survey the resulting family of methods [7]. Wong et al. show that FGSM training with random initialisation can approach PGD-based AT at a fraction of the cost, but may suffer from catastrophic overfitting [19]. Rice et al. identify robust overfitting, where robust test accuracy degrades late in training, and recommend early stopping on robust validation accuracy [20]. Later methods mitigate it by flattening the weight loss landscape [21], rebalancing the trainer–attacker game [22] or using self-distilled soft labels [23]. Our objective (Eq. (4)) is deliberately simple, a mix of clean and adversarial cross-entropy in the spirit of [2]; AdvProp [24] trains on the same mixture but uses separate Batch Normalisation statistics for the two input types, which we do not do. AT is also highly sensitive to basic settings such as weight decay and schedule [25], so we keep one untuned recipe for both networks.

C. The Accuracy-Robustness Trade-off

Tsipras et al. argue that robustness and accuracy can be fundamentally at odds [8], and TRADES makes the trade-off explicit by splitting robust error into natural and boundary error [9]. The clean-accuracy drop has been attributed to an overly complex decision boundary caused by insufficient learning of hard adversarial examples [26], motivating methods that limit the excessive margin AT builds along adversarial directions [10]. CURE selectively updates parts of the network and introduced the Natural-Robustness Ratio (NRR) [11]. Wang and Liu [12] measure perception consistency as the mean squared error (MSE) between the logits of a clean input and its perturbed version, separately for successfully defended and failed samples. For PGD-AT with ResNet-18 on CIFAR-10, failed samples have a smaller logit change than defended ones, i.e., failure cases are learned over-sufficiently, and they propose RPAT to make logits change smoothly along the perturbation. We reuse their measurement to ask whether the same pattern holds for a weaker AT objective and for a plain sequential network.

D. Architecture and Robustness

Su et al. benchmark 18 standardly trained ImageNet models, including VGG networks, and find that empirical robustness scales with the logarithm of classification error [27], i.e., robustness and clean accuracy are already linked across architectures. Bai et al. show that the reported robustness advantage of Transformers over CNNs largely disappears once both are trained with the same recipe [28]. Huang et al. study how depth and width should be allocated across residual stages for adversarially robust networks [13], and Peng et al. distil architectural design principles for robust CNNs from a large-scale study [14]. From a theoretical angle, architecture affects the Lipschitz constant and sensitivity of the model [29]. These works compare models mainly by absolute robust accuracy. We follow the controlled-recipe argument of [28], but for adversarially trained CNNs, and explicitly separate the contribution of clean accuracy.

III. Methodology

A. Threat Model and Adversarial Training

Let f_θ be a classifier and $J(\theta, x, y)$ the cross-entropy loss. Under the white-box threat model the adversary knows the architecture and weights and searches for a perturbation δ with $\|\delta\|_\infty \leq \epsilon$ that maximises the loss. Adversarial training solves [6]

$$\min_\theta \ \mathbb{E}_{(x,y)}\Big[\max_{\|\delta\|_\infty \leq \epsilon} J(\theta, x+\delta, y)\Big]. \tag{1}$$

Pixel values lie in $[0, 1]$; input normalisation is the first layer of each model, so all perturbation budgets are expressed in pixel units.

B. Attacks

FGSM takes one signed-gradient step of size ϵ [2]:

$$x_{\text{adv}} = \text{clip}_{[0,1]}\big(x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))\big). \quad (2)$$

PGD-K starts from a uniformly random point in the ϵ-ball $\mathcal{B}_\epsilon(x)$ and iterates [6]

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)\cap[0,1]^d}\big(x^t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x^t, y))\big). \quad (3)$$

We use PGD-7 with $\alpha = \epsilon/4$ and PGD-20 with $\alpha = 2/255$. AutoAttack (standard version: APGD-CE, APGD-T, FAB-T, Square) [18] is run at $\epsilon = 4/255$ and $8/255$ on the first 1,000 test images, on which we also report PGD-20 for a direct comparison.

C. Network Architectures

ResNet-18 [16] (11.22M parameters) stacks residual blocks with identity shortcuts; we use a 3×3 stem without initial max-pooling for 32×32 inputs. VGG-16 [17] (15.30M parameters) is a plain stack of 3×3 convolutions with Batch Normalisation and no shortcuts, followed by global average pooling and a classifier with two hidden layers of 512 units (Batch Normalisation, ReLU, dropout 0.5). The networks therefore differ in more than skip connections, which we take into account in Section V-D.

D. Mixed-Loss Adversarial Training

We minimise an equally weighted sum of clean and adversarial loss:

$$L = \tfrac{1}{2}J(\theta, x, y) + \tfrac{1}{2}J(\theta, x_{\text{adv}}, y), \quad (4)$$

where x_{adv} is generated for every mini-batch by PGD with 3 steps, $\epsilon = 4/255$ and $\alpha = 1/255$, with Batch Normalisation in inference mode while crafting the attack. Clean and adversarial inputs are then passed through the network in two separate forward passes, so each is normalised with its own batch statistics, while the running statistics are shared. The training budget is deliberately smaller than the largest evaluation budget, so that we can observe how robustness transfers to a budget not seen in training. Algorithm 1 lists the training steps; standard training uses the same loop with $L = J(\theta, x, y)$ and no attack.

E. Evaluation Protocol and Metrics

Every trained model is evaluated on the CIFAR-100 test set with the steps of Algorithm 2. We report (i) clean accuracy A_{clean}; (ii) robust accuracy A_{rob}, the top-1 accuracy on adversarial test images; (iii) the Macro F1-score over the 100 classes (the test set is class-balanced, so it closely tracks accuracy and is reported for completeness); (iv) the Natural-Robustness Ratio [11]

$$\text{NRR} = \frac{2\,A_{\text{clean}}\,A_{\text{rob}}}{A_{\text{clean}} + A_{\text{rob}}}, \quad (5)$$

the harmonic mean of the two accuracies; (v) the conditional robust accuracy

$$\rho = \Pr\big(f(x_{\text{adv}}) = y \,\big|\, f(x) = y\big), \qquad A_{\text{rob}} = A_{\text{clean}} \cdot \rho, \quad (6)$$

i.e., the fraction of correctly classified test images that remain correct under attack, or one minus the attack success rate on those images [27];[1] and (vi) perception consistency [12]: the mean logit MSE between each clean image and its PGD-20 version, computed separately for defended and failed samples among those classified correctly on clean data. We report the ratio $r = \text{MSE}_{\text{failed}}/\text{MSE}_{\text{defended}}$ (only when at least 30 samples are defended); $r < 1$ corresponds to the "failure cases are better learned" pattern of [12].

By Eq. (6), the gap in robust accuracy between the adversarially trained networks can be split exactly into a part due to clean accuracy and a part due to conditional robustness:

$$\Delta A_{\text{rob}} = \underbrace{\Delta A_{\text{clean}} \cdot \rho^{\text{VGG}}}_{\text{clean accuracy}} + \underbrace{A_{\text{clean}}^{\text{ResNet}} \cdot \Delta\rho}_{\text{conditional robustness}}. \quad (7)$$

The split depends on which network is used as reference; the reverse order changes the shares by at most 5 points (Table III).

Algorithm 1 Mixed-loss adversarial training (one seed)

Input: training set $\mathcal{D}$, validation set $\mathcal{D}_{\text{val}}$, network f_θ (ResNet-18 or VGG-16), epochs T=50, budget ϵ=4/255, step α=1/255, steps K=3
Output: final weights θ_T and best weights θ^*

for epoch $t = 1, \dots, T$ **do**
  for each mini-batch $(x, y) \subset \mathcal{D}$ of size 128 **do**
    $x \leftarrow$ random crop and horizontal flip of x
    switch Batch Normalisation to inference mode
    $x' \leftarrow \text{clip}_{[0,1]}(x + u)$, $u \sim \mathcal{U}(-\epsilon, \epsilon)^d$
    for $k = 1, \dots, K$ **do**
      $x' \leftarrow \Pi_{\mathcal{B}_\epsilon(x)\cap[0,1]^d}\big(x' + \alpha\,\text{sign}(\nabla_{x'} J(\theta, x', y))\big)$
    switch Batch Normalisation to training mode
    $L \leftarrow \frac{1}{2}J(\theta, x, y) + \frac{1}{2}J(\theta, x', y)$ ▷ Eq. (4)
    $\theta \leftarrow$ SGD step on L (momentum 0.9, wd 5×10^{-4})
  divide the learning rate by 10 if $t \in \{25, 37\}$
  $a_t \leftarrow$ PGD-10 (ϵ=8/255) accuracy on $\mathcal{D}_{\text{val}}$
  if a_t is the highest so far **then** $\theta^* \leftarrow \theta$
return θ_T (main result) and θ^* (secondary)

Algorithm 2 Evaluation and gap decomposition

Input: trained models f (2 networks × 2 objectives × 3 seeds), test set $\mathcal{D}_{\text{test}}$, budgets $\mathcal{E} = \{2, 4, 8\}/255$

for each model f **do**
  A_{clean}, Macro F1 on $\mathcal{D}_{\text{test}}$; $\mathcal{C} \leftarrow \{i : f(x_i) = y_i\}$
  for each $\epsilon \in \mathcal{E}$ **do**
    for attack ∈ {FGSM, PGD-7, PGD-20} **do**
      $x_i^{\text{adv}} \leftarrow \text{attack}(f, x_i, y_i, \epsilon)$; $A_{\text{rob}} \leftarrow$ accuracy
    $\rho \leftarrow |\{i \in \mathcal{C} : f(x_i^{\text{adv}}) = y_i\}|/|\mathcal{C}|$ (PGD-20)
    NRR by Eq. (5); logit-MSE ratio r over $\mathcal{C}$
  $A_{\text{AA}} \leftarrow$ AutoAttack on first 1,000 test images, $\epsilon \in \{4, 8\}/255$
  check: PGD-20 at ϵ=64/255 gives 0%; $A_{\text{AA}} \le A_{\text{PGD-20}}$ on the same images
average all metrics over seeds (mean ± std)
for each ϵ: split ΔA_{rob} by Eq. (7)

[1] The identity holds when inputs misclassified on clean data remain misclassified under attack, which is the case for nearly all samples in our runs.

TABLE I

RESULTS ON CIFAR-100 AT $\epsilon = 8/255$ (ℓ_∞), I.E., TWICE THE AT TRAINING BUDGET; RESULTS AT THE TRAINING BUDGET ARE IN TABLE II. FINAL CHECKPOINT, MEAN $\pm$ STD OVER 3 SEEDS. ACCURACIES IN %, MACRO F1 IN $[0, 1]$.

Model	Clean Acc.	Clean F1	FGSM	PGD-7	PGD-20	AutoAttack
ResNet-18 (standard)	74.96±0.03	0.750	6.68±0.47	0.01±0.01	0.00±0.00	0.00±0.00
ResNet-18 (mixed AT)	69.81±0.04	0.697	17.37±0.12	10.27±0.30	7.10±0.16	5.77±0.31
VGG-16 (standard)	61.84±0.15	0.622	6.19±0.43	0.22±0.09	0.07±0.06	0.00±0.00
VGG-16 (mixed AT)	57.90±0.64	0.577	14.99±0.27	8.86±0.14	6.03±0.16	4.13±0.12

IV. EXPERIMENTAL SETUP

Data and training. CIFAR-100 (50,000 training and 10,000 test images, 100 classes). We hold out 1,000 training images as a validation set. All models use SGD with momentum 0.9, batch size 128, weight decay 5×10^{-4} and initial learning rate 0.1, the optimiser settings of [12], for 50 epochs, with the learning rate divided by 10 after epochs 25 and 37. Data augmentation is random cropping with 4-pixel padding and horizontal flipping. Standard and adversarially trained models share all hyperparameters.

Checkpoints. We report the final (last-epoch) checkpoint as our main result. As a secondary result we also report the best checkpoint, selected by clean validation accuracy for standard models and by PGD-10 ($\epsilon = 8/255$) validation accuracy for AT models, as in [12]. Section V-E explains why we do not use the best checkpoint as the main one.

Seeds and hardware. Each configuration is trained with 3 random seeds; we report mean $\pm$ standard deviation. Training ran on NVIDIA Tesla T4 GPUs in Google Colab and Kaggle (PyTorch 2.11 on Kaggle). Runs interrupted by session limits were resumed from the last saved epoch, restoring model, optimiser and scheduler state.

Sanity checks. Following standard practice [4], [18], we verify that PGD-20 with a very large budget ($\epsilon = 64/255$) drives accuracy of every model to 0%, and that AutoAttack is not weaker than PGD.

V. RESULTS

A. Overall Results and Evaluation Validity

Table I summarises all models at $\epsilon = 8/255$. The attacks are ordered as expected (FGSM $>$ PGD-7 $>$ PGD-20 $>$ AutoAttack), the large-budget sanity check reaches 0% for every model, and inside AutoAttack the black-box Square attack finds no additional adversarial examples beyond APGD (FAB-T adds a single example in one run). These checks indicate that the measured robustness is not caused by gradient masking. At the training budget $4/255$, AutoAttack is only 1.3–1.6 points below PGD-20 on the same 1,000 images (ResNet-18: 27.37 vs. 28.63%; VGG-16: 20.93 vs. 22.50%).

B. Standard Training

Standard training gives clean accuracies of 74.96% (ResNet-18) and 61.84% (VGG-16), but both models are almost defenceless. Already at $\epsilon = 2/255$, PGD-20 reduces accuracy to 3.06% and 4.23%, and at $\epsilon = 4/255$ to at most 0.5% (Table II, Fig. 2). At small budgets the standard VGG-16 is slightly more robust than the standard ResNet-18: 6.8% versus 4.1% of its correctly classified images survive PGD-20 at $\epsilon = 2/255$. Higher clean accuracy does not, by itself, imply higher robustness.

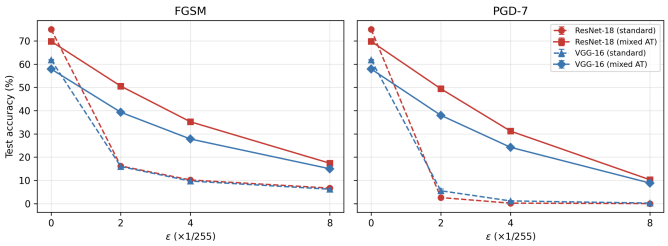


Fig. 2. Test accuracy of standard (dashed) and adversarially trained (solid) ResNet-18 and VGG-16 on CIFAR-100 versus perturbation budget ϵ, under FGSM (left) and PGD-7 (right). Final checkpoint, mean over 3 seeds; error bars (std) are smaller than the markers.

TABLE II

ACCURACY (%) UNDER PGD-20 AT DIFFERENT BUDGETS, FINAL CHECKPOINT. NRR FROM CLEAN AND PGD-20 ACCURACY (EQ. (5)); ρ: CONDITIONAL ROBUST ACCURACY (EQ. (6)).

Model	ϵ	PGD-20	NRR	ρ
ResNet-18 (std.)	2/255	3.06±0.20	5.9	4.1±0.3
VGG-16 (std.)	2/255	4.23±0.36	7.9	6.8±0.6
ResNet-18 (AT)	2/255	48.29±0.33	57.1	69.2±0.5
	4/255	28.17±0.37	40.1	40.3±0.5
	8/255	7.10±0.16	12.9	10.2±0.2
VGG-16 (AT)	2/255	37.05±0.63	45.2	63.9±0.4
	4/255	21.84±0.19	31.7	37.7±0.1
	8/255	6.03±0.16	10.9	10.4±0.4

C. Effect of Mixed-Loss Adversarial Training

AT lowers clean accuracy by 5.15 points for ResNet-18 ($74.96 \to 69.81\%$) and by 3.94 points for VGG-16 ($61.84 \to 57.90\%$). In return, PGD-20 accuracy at the training budget $\epsilon = 4/255$ rises from 0.07% to 28.17% for ResNet-18 and from 0.50% to 21.84% for VGG-16, and at $\epsilon = 2/255$ to 48.29% and 37.05% (Table II). The protection transfers poorly to twice the training budget: at $\epsilon = 8/255$ PGD-20 accuracy is only 7.10% and 6.03% (AutoAttack: 5.77% and 4.13%), and only about 10% of correctly classified images survive. Robustness obtained with this objective is therefore largely specific to the budget used in training.

TABLE III
DECOMPOSITION (EQ. (7)) OF THE PGD-20 GAP BETWEEN ADVERSARIALLY TRAINED RESNET-18 AND VGG-16 (POINTS), FINAL CHECKPOINT, MEAN ± STD OVER SEED PAIRS. REVERSE REFERENCE ORDER: 73/76/114%.

ϵ	Gap	Clean-acc. part	Conditional part
2/255	11.24±0.95	7.61±0.37 (68%)	3.65±0.60
4/255	6.33±0.55	4.49±0.26 (71%)	1.84±0.30
8/255	1.07±0.03	1.24±0.11 (116%)	−0.18±0.09

D. Architecture: Robustness or Clean Accuracy?

After AT, ResNet-18 has higher robust accuracy than VGG-16 at every budget (Table II). Table III splits this gap with Eq. (7). At $\epsilon = 2/255$ and $4/255$, about 70% of the gap comes from ResNet-18's higher clean accuracy and about 30% from a genuinely higher survival rate of correctly classified images ($\Delta\rho = 5.3$ and 2.6 points). At $\epsilon = 8/255$ the two networks have the same conditional robustness (10.2% vs. 10.4%), and the whole gap is explained by clean accuracy. Thus residual connections, or the other differences between the two networks, mainly make adversarial training reach a higher clean accuracy. They make the resulting model only moderately harder to attack at small budgets, and not at all at $\epsilon = 8/255$. Our VGG-16 is below common CIFAR-100 implementations, so we ran a control (3 seeds, same recipe) in which its two-layer classifier head is replaced by a single linear layer (VGG-16-lin, 14.77M parameters). The head raises clean accuracy to 69.14% (standard) and 64.12% (AT) but leaves conditional robustness essentially unchanged ($\rho = 36.5$ vs. 37.7% at $4/255$); PGD-20 accuracy at $4/255$ rises from 21.84 to 23.43% (AutoAttack 22.77%). Against this stronger baseline the clean-accuracy share of ResNet-18's advantage is 58%, 45% and 41% at $\epsilon = 2, 4, 8/255$, and ResNet-18 keeps a small conditional advantage at every budget ($\Delta\rho = 3.9$, 3.8, 1.2 points). The exact share thus depends on how well the weaker baseline is trained, but clean accuracy remains a major part of the gap, and the head affects robust accuracy only through clean accuracy. Because the networks also differ in depth, parameter count and classifier head, we cannot attribute the remaining conditional advantage to skip connections specifically; controlled variants (e.g., VGG-16 with shortcuts) would be needed [13], [14].

E. Checkpoint Selection and Robust Overfitting

Table IV compares the best and final checkpoints of the AT models. The best checkpoint is slightly more robust (PGD-20 8.10% vs. 7.10% for ResNet-18, 6.46% vs. 6.03% for VGG-16; AutoAttack 0.7–0.8 points higher), a mild form of robust overfitting [20]. It is, however, much less stable for VGG-16: its clean accuracy varies by ± 2.54 points across seeds, compared with ± 0.64 for the final checkpoint, and it is on average 4.8 points lower. The cause is the selection signal: following [12] we select at $\epsilon = 8/255$, above the training budget, where robust validation accuracy is only about 7% on 1,000 images, so checkpoints that differ by one or two correctly classified images are ranked differently, and an early checkpoint with low clean accuracy can be selected. Measured with best checkpoints, the clean-accuracy cost of AT for VGG-16 would be 8.3 points ($61.41 \rightarrow 53.07\%$) instead of 3.9, overstating the trade-off by a factor of two. We therefore use the final checkpoint for the main comparison and recommend that studies selecting "best" checkpoints report the validation-set size, the selection budget and the variance this choice introduces.

TABLE IV
BEST VS. FINAL CHECKPOINT OF THE AT MODELS AT $\epsilon = 8/255$.

Model	Ckpt	Clean	PGD-20	AutoAttack
ResNet-18 (AT)	best	68.29±0.59	8.10±0.16	6.47±0.17
	final	69.81±0.04	7.10±0.16	5.77±0.31
VGG-16 (AT)	best	53.07±2.54	6.46±0.32	4.90±0.41
	final	57.90±0.64	6.03±0.16	4.13±0.12

TABLE V
PERCEPTION CONSISTENCY: RATIO $r = \mathrm{MSE}_{\mathrm{failed}}/\mathrm{MSE}_{\mathrm{defended}}$ OF LOGIT MSE BETWEEN CLEAN AND PGD-20 INPUTS, FINAL CHECKPOINT. $r < 1$ WOULD INDICATE THAT FAILURE CASES ARE LEARNED BETTER [12]. "–": FEWER THAN 30 DEFENDED SAMPLES.

Model	$\epsilon = 2/255$	$4/255$	$8/255$
ResNet-18 (std.)	2.736±0.161	–	–
VGG-16 (std.)	14.343±0.929	26.936±5.787	–
ResNet-18 (AT)	1.519±0.005	1.790±0.009	2.656±0.051
VGG-16 (AT)	4.576±0.167	5.917±0.241	8.224±0.729

F. Perception Consistency

Table V reports the ratio r of logit MSE between failed and defended samples. For standard models failed samples change their logits far more than defended ones ($r = 2.7$ and 14.3 at $\epsilon = 2/255$). AT reduces the ratio to 1.5 and 4.6, and ResNet-18 is closer to $r = 1$ at every budget. However, r stays above 1 in all our settings: the pattern reported by Wang and Liu [12] for PGD-AT, where failure cases have *smaller* logit change than defended ones, does not appear under our objective. A likely reason is that our AT is weaker (budget $4/255$, 3 PGD steps, half of the loss on clean data), so the over-sufficient learning of failure cases that they describe has not yet occurred. The strength of this effect thus depends on both the AT objective and the architecture. Absolute MSE values are not comparable across networks, because the logit scales differ; only ratios within one model are compared.

G. Limitations

(1) Our AT objective is weaker than standard PGD-AT at $\epsilon = 8/255$. Published PGD-AT ResNet-18 models on CIFAR-100 reach 56.6% clean and 25.0% AutoAttack accuracy at $8/255$ [12], and the PreActResNet-18 of [20] 53.8% and 19.0% [15]; our ResNet-18 instead trades robustness at $8/255$ (5.8%) for higher clean accuracy (69.8%), so conclusions about $8/255$ should not be extrapolated to standard PGD-AT. (2) Because of our compute budget (T4 GPUs, 50 epochs,

weak AT), absolute accuracies are below published PGD-AT results and VGG-16 is below common implementations; robustness comparisons are known to be sensitive to such training details [25], [28], and our VGG-16-lin control shows that the clean-accuracy share depends on baseline strength (71% vs. 45% at $4/255$). (3) AutoAttack is run on 1,000 test images, giving a standard error of 0.7–1.4 points. (4) We use one dataset and two architectures, and the architectures differ in more than skip connections. (5) Results are averaged over 3 seeds; small differences (< 1 point) should be read with caution.

VI. Conclusion

We compared ResNet-18 and VGG-16 on CIFAR-100 under an identical recipe, with and without mixed-loss AT, over three seeds. AT at $\epsilon = 4/255$ gives substantial robustness at that budget for a clean-accuracy cost of 3.9–5.2 points, but little at $\epsilon = 8/255$. ResNet-18 is more robust in absolute terms, yet about 70% of its advantage at small budgets, and all of it at $\epsilon = 8/255$, follows from its higher clean accuracy; against a better-trained VGG-16 control this share is still 41–58%. We therefore suggest that architecture comparisons in adversarial robustness report conditional robust accuracy alongside absolute robust accuracy. We also showed that best-checkpoint selection on a small robust validation set can distort the measured trade-off, and that the perception pattern reported in [12] depends on the strength of adversarial training. Future work includes standard PGD-AT at $\epsilon = 8/255$, controlled architecture ablations, additional datasets, and testing whether trade-off methods such as RPAT [12] and CURE [11] change the decomposition reported here.

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